DO ALL TRANSIT RIDERS MINIMIZE TRAVEL TIME? EVIDENCE OF BEHAVIORAL HETEROGENEITY FROM LATENT CLASS ANALYSIS AND MNL RE-RANKING OF 1.3 MILLION SMART CARD TRIPS IN SEOUL

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ABSTRACT

Transit routing applications serve millions of travelers daily, yet they uniformly assume homogeneous preferences, potentially misserving demographic groups with distinct behavioral needs. This study investigates whether theory-grounded discrete choice models can match the predictive accuracy of machine learning for transit route choice, and whether the behavioral insights they yield can improve operational routing systems. Using 1,320,030 trip chains from Seoul's Transportation Card Data matched to OpenTripPlanner (OTP) alternatives, we compare four model families—Multinomial Logit (MNL), Mixed Logit, Latent Class (LC, K=3), and LightGBM—on an identical hold-out set of 82,351 chains. The LC model achieves the highest Hit Rate (67.4%), outperforming LightGBM (66.2%) with an order of magnitude fewer parameters. Three behaviorally distinct classes emerge—including an Accessibility-Priority segment (8.7%, predominantly elderly) that values longer seated rides over transfers—yet administrative card types only partially predict class membership (Cramér's V = 0.685). Variance-standardized MNL coefficients and SHAP importance agree on the dominant factors, walking time and transfers (Spearman ρ = 0.8). On the practical side, traditional calibration of OTP parameters encounters a ceiling of approximately one percentage point, whereas MNL-based re-ranking of existing alternatives delivers +1.89 percentage points at negligible computational cost, with a 10.4-to-1 win ratio over the router when the two disagree. These results suggest that behavioral theory retains a structural advantage over black-box prediction in choice settings, and that lightweight model-based post-processing can meaningfully upgrade operational transit routing.

Keywords: Transit route choice, Behavioral heterogeneity, Latent Class model, Smart card big data, MNL re-ranking, OpenTripPlanner, RAPTOR algorithm

INTRODUCTION

Every day, millions of transit travelers consult routing applications—Google Maps, Naver Maps, or open-source engines such as OpenTripPlanner (OTP)—to decide how to get from origin to destination. These platforms uniformly adopt a deterministic paradigm: they minimize a predefined generalized cost and present the result as the single "best" route, implicitly assuming that all users weigh walking, riding, and transferring in the same way. Empirical evidence, however, paints a different picture. Elderly passengers may tolerate longer rides to avoid stairway transfers; wheelchair users may detour to reach an elevator; commuters may prioritize speed above all else. If these preferences are systematic rather than random, a one-size-fits-all recommendation will systematically mis-serve certain groups—a concern with direct equity implications as urban populations age.

Seoul provides an ideal setting for investigating these questions. Its Transportation Card system records over seven million tap-in/tap-out transactions daily across all bus and subway services, enabling population-scale revealed-preference observation of route choice. Five administratively defined card types—general, elderly, disabled, youth, and children—permit demographic segmentation without survey recruitment, and OpenTripPlanner's RAPTOR algorithm (5) can systematically generate competing alternatives for each observed trip. Prior work using Seoul smart card data (9) calibrated a transit route choice model but considered only a single user type; the present study extends this to four model families, five user types, and 1.32 million trip chains.

Despite a growing literature on transit route choice, three substantive gaps remain. First, most studies estimate a single discrete choice specification and report its fit, without testing whether the structure imposed by random utility theory helps or hinders prediction relative to flexible machine learning alternatives such as gradient boosting. Second, neither Mixed Logit's continuous coefficient distributions nor observed demographic segmentation can discover data-driven behavioral clusters that cut across administrative card-type categories—a 70-year-old commuter who walks briskly differs fundamentally from a frail peer who avoids stairs, yet both carry the same "elderly" smart card. Third, route choice research overwhelmingly concludes at the point of parameter estimation; whether those parameters can improve an operational routing system—and if so, by how much—is seldom examined.

The present study pursues four objectives. First, we develop a multi-level route similarity framework with 10 metrics organized into four hierarchical levels. Second, we estimate and compare four model families—MNL, Mixed Logit, Latent Class (with concomitant variables), and LightGBM—on an identical hold-out test set drawn from 1.32 million trip chains. Third, we characterize the latent behavioral segments and examine how they align with administrative card-type categories. Fourth, we close the loop between estimation and practice by demonstrating that MNL-based re-ranking of router-generated alternatives outperforms traditional parameter calibration at negligible computational cost.

LITERATURE REVIEW

TRANSIT ROUTE CHOICE MODELS

The theoretical foundation for route choice modeling rests on random utility maximization (3), under which a traveler selects the alternative with the highest perceived utility. Prato (12) identifies three persistent challenges: generating plausible alternatives, defining attributes behaviorally, and selecting a model structure. The MNL model (10) remains the workhorse owing to its tractability, though its IIA limitation has motivated richer structures. Train (14) demonstrated that Mixed Logit captures unobserved taste heterogeneity through continuous coefficient distributions, while Greene and Hensher (6) showed that Latent Class models segment the population into discrete behavioral groups linked to observable characteristics via concomitant variables. Large-scale empirical applications remain scarce: Kim et al. (9) calibrated a transit route choice model using Seoul smart card data but considered a single user type, Arriagada et al. (1) estimated a learning-based model on ~100,000 trips in Santiago, and Arriagada et al. (2) later revealed route choice strategy heterogeneity using smart card data in the same network.

TRANSFER PENALTY AND USER HETEROGENEITY

Jara-Díaz et al. (7), synthesizing data from five countries, reported pure transfer penalties of 13–18 minutes. For Seoul, Yoo (15) estimated an average penalty of 11.24 minutes but did not disaggregate by demographics. A growing body of work documents systematic differences across traveler groups: Chen et al. (4) found significant preference heterogeneity among elderly riders in China, and Peng et al. (11) reported that older rail passengers are 10.2 percentage points less likely to choose express services. These studies examine single segments in isolation; no prior work has simultaneously compared behavior across five user types using four model families—an omission this paper rectifies.

DIFFERENTIATION FROM PRIOR RESEARCH

Table 1 positions the present study relative to the most relevant prior work. The key distinctions are the data scale (1.32 million chains), model breadth (four families), multi-level similarity framework, and closed-loop calibration analysis.

Criterion	(9)	(15)	(1)	This Study (2025)
Data Scale	~10K	~1K	~100K	1,320K
Similarity	Single	Single	Single	4-Level/10
Models	MNL	MNL	MNL	4 Families
User Types	1	1	1	5 + LC
ML Benchmark	×	×	×	LightGBM
Calibration	×	×	×	θ + Re-rank

Table 1. Comparison with Prior Research

DATA AND METHODOLOGY

DATA OVERVIEW

The empirical foundation is Seoul's Transportation Card Data (TCD), which records tap-in and tap-out transactions for every transit trip. TCD captures subway-to-subway transfers that older smart card formats miss, yielding a more complete picture of multimodal trip chains. We analyze a single representative weekday, February 20, 2025 (Thursday). After removing trips with no valid OTP alternative, the dataset comprises 1,320,030 trip chains paired with 4,936,864 chain–alternative observations (3.74 alternatives per chain on average).

Chain Type	Chains	Share	Avg. Alternatives
Bus Only	445,554	33.8%	3.51
Subway Only	720,692	54.6%	3.82
Mixed (Bus+Subway)	153,784	11.7%	4.01
Total	1,320,030	100%	3.74

Table 2. Trip Chain Composition

For model estimation, chains with at least two alternatives (411,754) are split 80/20 into training (329,403) and test (82,351) sets, stratified by user type.

User Type	All Chains	Share	Train	Test	Test HR Range
General	1,044,091	79.1%	271,843	67,961	64.8–66.4%
Elderly	177,303	13.4%	24,132	6,033	71.6–77.8%
Youth	49,772	3.8%	19,725	4,931	64.9–67.0%
Disabled	38,281	2.9%	9,113	2,278	70.3–71.5%
Children	10,583	0.8%	4,590	1,148	64.9–69.4%

Table 3. User Type Distribution and Test Hit Rate Range Across Models

MULTI-LEVEL SIMILARITY FRAMEWORK

Rather than relying on a single similarity metric—a limitation noted by Tomhave and Khani (13) in their multi-criteria transit route choice study—we construct a hierarchical framework with four levels and ten metrics (Table 4). Level 1 (Exact Match) is the strictest criterion, requiring complete identity of route sequences, mode sequences, and boarding/alighting stops. Level 2 (Structural Similarity) relaxes this to sequence-level measures using Longest Common Subsequence (LCS) similarity. Level 3 (Stop-based Similarity) operates at the individual stop level via Jaccard indices. Level 4 (Temporal Similarity) captures travel-time correspondence. The composite index is:

$$\begin{aligned} sim_{\text{total}} &= 0.30 \cdot sim_{\text{route\_seq}} + 0.20 \cdot sim_{\text{mode\_seq}} + 0.20 \cdot sim_{\text{jaccard\_full}} \\ &\quad + 0.15 \cdot sim_{\text{lcs}} + 0.15 \cdot sim_{\text{time}} \end{aligned}$$

Level	Metric	Description	Weight
1. Exact	exact_match	Route + mode + stop identity	—
2. Structural	sim_route_seq	Route sequence LCS similarity	0.30
	sim_mode_seq	Mode sequence LCS similarity	0.20
	transfer_match	Transfer count match (binary)	—
3. Stop-based	sim_jaccard_board	Boarding stop Jaccard index	—
	sim_jaccard_alight	Alighting stop Jaccard index	—
	sim_jaccard_full	All-stop Jaccard index	0.20
	sim_lcs	Stop sequence LCS similarity	0.15
4. Temporal	sim_time	Travel time ratio similarity	0.15
Composite	sim_total	Weighted sum of above	1.00

Table 4. Multi-Level Similarity Framework (10 Metrics)

Level 1 (Exact Match) requires complete identity of route, mode, and stop sequences. Level 2 (Structural Similarity) uses normalized LCS similarity: $2 \cdot \text{LCS}(A, B) / (|A| + |B|)$. Level 3 (Stop-based Similarity) applies the Jaccard index $|S_{\text{obs}} \cap S_{\text{alt}}| / |S_{\text{obs}} \cup S_{\text{alt}}|$ to boarding, alighting, and all traversed stops, complemented by stop-sequence LCS to preserve order. Level 4 (Temporal Similarity) is defined as $sim_{\text{time}} = 1 - \min(1, |T_{\text{obs}} - T_{\text{alt}}| / \max(T_{\text{obs}}, T_{\text{alt}}))$. Because TCD records intermediate stops only for bus legs, subway segments use boarding and alighting stops alone—a limitation the Jaccard formulation handles gracefully.

MODEL SPECIFICATIONS

Multinomial Logit (MNL)

The utility for alternative j in individual n's choice set is:

$$\begin{aligned} V_j &= \beta_{\text{ride}} T_{\text{ride}} + \beta_{\text{walk}} T_{\text{walk}} + \beta_{\text{transfer}} N_{\text{transfer}} + \beta_{\text{subway}} D_{\text{subway}} \\ &\quad + \beta_{\text{peak}\cdot\text{ride}} (D_{\text{peak}} \times T_{\text{ride}}) + \beta_{\text{peak}\cdot\text{xfer}} (D_{\text{peak}} \times N_{\text{transfer}}) \end{aligned}$$

where T_{ride} is in-vehicle time (min), T_{walk} is total walking time (min), N_{transfer} is transfer count, D_{subway} indicates subway inclusion, and D_{peak} flags peak-hour departures (07–09h, 18–20h). Waiting time was excluded after preliminary estimation revealed multicollinearity and a counter-theoretic positive coefficient.

Mixed Logit

To accommodate unobserved taste heterogeneity, T_{walk}, N_{transfer}, and D_{subway} are specified as random coefficients, $\beta_k \sim N(\mu_k, \sigma_k)$, while T_{ride} and peak interactions remain fixed. Significant σ_k values formally reject MNL's homogeneity assumption.

Latent Class Model

Following Greene and Hensher (6), a K-class model assigns each class its own utility parameters. Peak interactions are dropped after Mixed Logit reveals them non-significant once heterogeneity is controlled. Class membership is modeled as a function of concomitant variables $Z_n = [D_{\text{children}}, D_{\text{youth}}, D_{\text{elderly}}, D_{\text{disabled}}, D_{\text{peak}}]$.

LightGBM Benchmark

As a non-parametric benchmark, LightGBM (8) classifiers predict which alternative was chosen. Two configurations are tested: a Core model (same four variables as MNL) and a Full model adding D_{peak}, user_type, and chain composition.

RESULTS

ROUTE SIMILARITY ANALYSIS

Before model estimation, we assess how well OTP's alternatives match observed behavior. The Best Match Exact Match rate across 1.32 million chains is 28.66%—the share for which at least one alternative perfectly reproduces the observed trip. The OTP 1st Rank rate is 20.34%, capturing what a traveler would experience following the router's top recommendation. The 8.3 percentage point gap reveals the room for improvement through better ranking.

Metric	Method	Mean	Median	Std. Dev.
Exact Match	Best Match	28.66%	—	—
Exact Match	OTP 1st Rank	20.34%	—	—
sim_total	Best Match	0.626	0.668	0.283
sim_total	OTP 1st Rank	0.537	0.562	0.312
sim_total ≥ 0.7	Best Match	42.5%	—	—

Table 5. Route Similarity Summary (N = 1,320,030)

Exact matching is heavily concentrated in direct trips (45.6% for zero-transfer chains) and drops sharply for one-transfer (3.7%) and two-or-more-transfer trips (<0.3%), reflecting the combinatorial explosion of route possibilities as journey complexity increases.

Figure 1 reveals a bimodal distribution: the largest concentration falls in the 90–100% bin (23.1%), while two nearly equal secondary modes appear at 30–40% and 40–50% (14.8% each), separated by a valley at 50–60% (5.1%). Overall, 42.5% of chains achieve sim_total ≥ 0.70, confirming that OTP generates at least one behaviorally close alternative for a substantial share of trips, while the remaining 57.5% highlights the scope for model-based re-ranking.

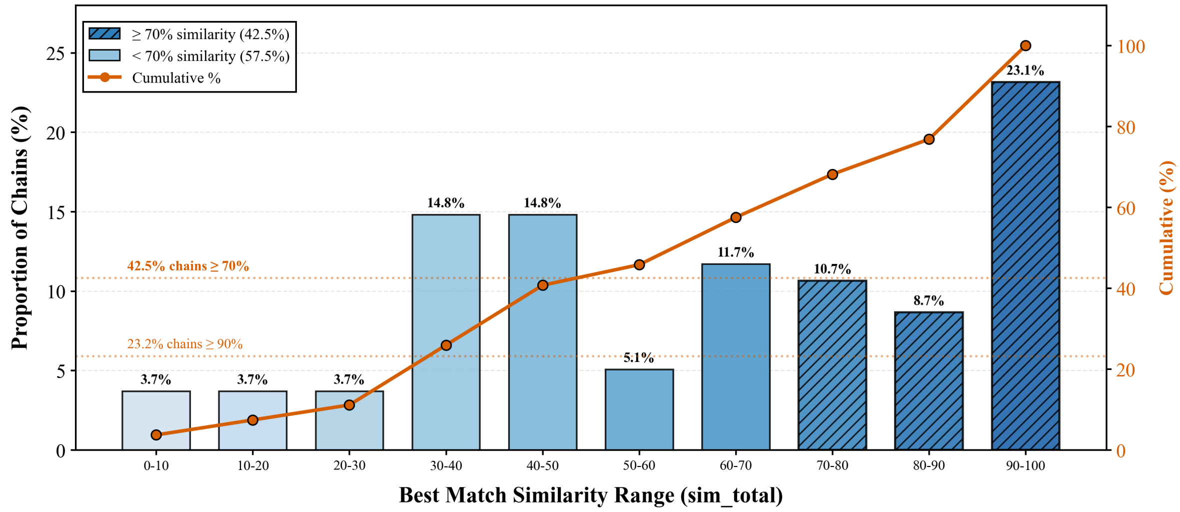

Figure 1. Distribution of Best-Match Similarity (sim_total) Across 1,320,030 Trip Chains

MODEL ESTIMATION RESULTS

MNL: Pooled and User-Type-Specific Results

Table 6 reports MNL estimates for the pooled sample and four user-type-specific models. All coefficients carry expected signs—negative for ride time, walking time, and transfers; positive for subway inclusion—with one striking exception.

Parameter	Pooled	General	Youth	Elderly	Disabled
β_{ride}	−0.034	−0.045	−0.078	+0.108	+0.028
β_{walk}	−0.776	−0.767	−0.880	−0.785	−0.840
$\beta_{transfer}$	−3.413	−3.322	−3.941	−4.290	−3.600
β_{subway}	+2.394	+2.161	+2.376	+4.641	+2.715
Walk Weight	22.7×	16.9×	11.2×	7.3×	30.2×
Transfer (min)	100.0	73.3	50.3	39.6	129.6
ρ^2	0.469	0.459	0.486	0.582	0.515
Hit Rate	62.9%	62.1%	63.8%	78.5%	73.0%

Table 6. MNL Parameter Estimates by User Type. All $p < 0.001$ except Disabled β_{ride} ($p < 0.01$)

The most notable result in Table 6 is the positive β_{ride} for elderly users (+0.108, t = 17.1). Under the standard assumption of time minimization, this coefficient should be negative; its reversal suggests that elderly travelers prefer longer in-vehicle times, likely because longer subway rides afford a seated, climate-controlled journey free of stair-climbing and crowd-navigation. Consistent with this interpretation, their subway coefficient (β_{subway} = +4.641) is more than double that of general users (+2.161).

At the other end of the spectrum, disabled users exhibit the highest walking weight (30.2×), implying that each minute of walking is perceived as equivalent to roughly half an hour of in-vehicle travel. Their transfer penalty of 129.6 minutes—nearly twice that of general users (73.3 min)—substantially exceeds the 13–18-minute range in Jara-Díaz et al.'s (7) international meta-analysis, suggesting that population-average estimates may severely understate the disutility faced by mobility-impaired subgroups.

Mixed Logit Results

Mixed Logit estimation ($\rho^2 = 0.464$) yields highly significant standard deviations for all three random coefficients ($p < 0.001$), with the walking weight spanning a 95% interval of 14.0× to 36.3× (LRT ≈ 1,040, df = 3, $p < 0.001$). Both peak interaction terms become non-significant once heterogeneity is accommodated ($p = 0.33$ and 0.86), suggesting that apparent peak effects in MNL were proxies for unobserved individual variation. The elderly subgroup harbors the widest within-group heterogeneity: their transfer penalty 95% range spans 7.9 to 76.7 minutes, confirming that the "elderly" card-type label conceals enormous behavioral diversity.

Latent Class Results (K = 3)

Parameter	Class 1 (8.7%)	Class 2 (77.2%)	Class 3 (14.0%)	Walk Wt.	Xfer (min)
β_{ride}	+0.252	−0.077	+1.166	—	—
β_{walk}	−1.123	−0.730	−10.000	$9.5\times^{2}$	—
β_{transfer}	−10.000	−3.272	−10.000	—	42.5^{2}
β_{subway}	+8.120	+2.022	+10.000	—	—

Table 7. Latent Class Utility Parameters. ^{2} = Class 2 only. All $p<0.001$

The BIC-optimal three-class solution reveals a strikingly interpretable segmentation. Class 1, "Accessibility-Priority Travelers" (8.7%), is characterized by positive ride-time valuation (+0.252), boundary-level transfer aversion ($\beta_{\text{transfer}} = -10.0$), and dominant subway preference ($\beta_{\text{subway}} = +8.12$). The concomitant variable for elderly card type yields an odds ratio of 22.0—yet only 61% of elderly users map to this class; the remaining 39% split between Mainstream (26%) and Extreme (13%) classes. The resulting Cramér's V of 0.685 indicates that card type and behavioral class are strongly correlated but far from identical—administrative demographics are a useful starting point for personalization, not an endpoint.

Class 2, the "Mainstream Travelers" (77.2%), exhibits rational trade-offs with a walking weight of 9.5× and transfer penalty of 42.5 minutes. Class 3, "Extreme Subway Loyalists" (14.0%), has every coefficient at or near its estimation boundary. The model achieves $\rho^2 = 0.474$ with excellent generalization: training and test Hit Rates (67.67% vs. 67.43%) differ by 0.24 percentage points, ruling out overfitting.

MULTI-MODEL COMPARISON

Model	Parameters	Hit Rate	Mean Prob	ρ^2
LC (K=3)	24	67.4%	60.3%	0.474
LightGBM Full	~313 trees	66.2%	40.4%	—
MNL (Pooled)	6	66.0%	60.0%	0.469
Mixed Logit	9	65.9%	60.7%	0.464
LightGBM Core	~596 trees	65.8%	43.5%	—

Table 8. Model Performance Comparison (Test Set: 82,351 Chains)

The Latent Class model leads at 67.4%, followed by LightGBM Full (66.2%), MNL (66.0%), Mixed Logit (65.9%), and LightGBM Core (65.8%). That a 24-parameter economic model outperforms a gradient boosting ensemble demonstrates that choice-set structure—the constraint that exactly one alternative is chosen from a defined set—carries substantial predictive information that LightGBM forfeits by treating alternatives independently. Equally striking is plain MNL, whose 66.0% represents 99.7% of LightGBM Full's performance with just six parameters. The

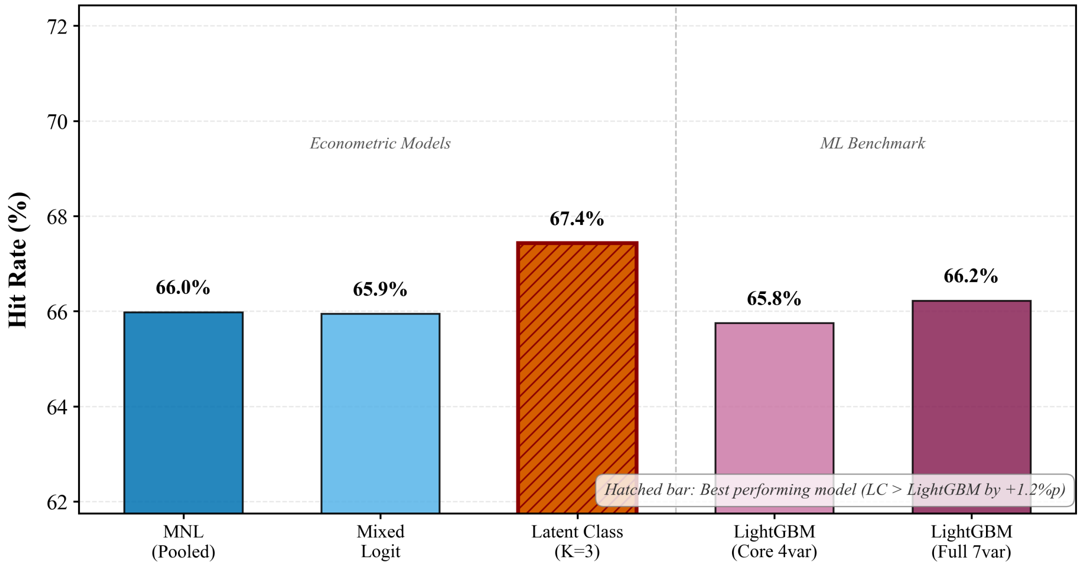

Figure 2. Model Hit Rate Comparison on the Hold-Out Test Set

tight 1.6 percentage point band suggests a natural accuracy ceiling imposed by unobserved factors.

Cross-Validation via SHAP

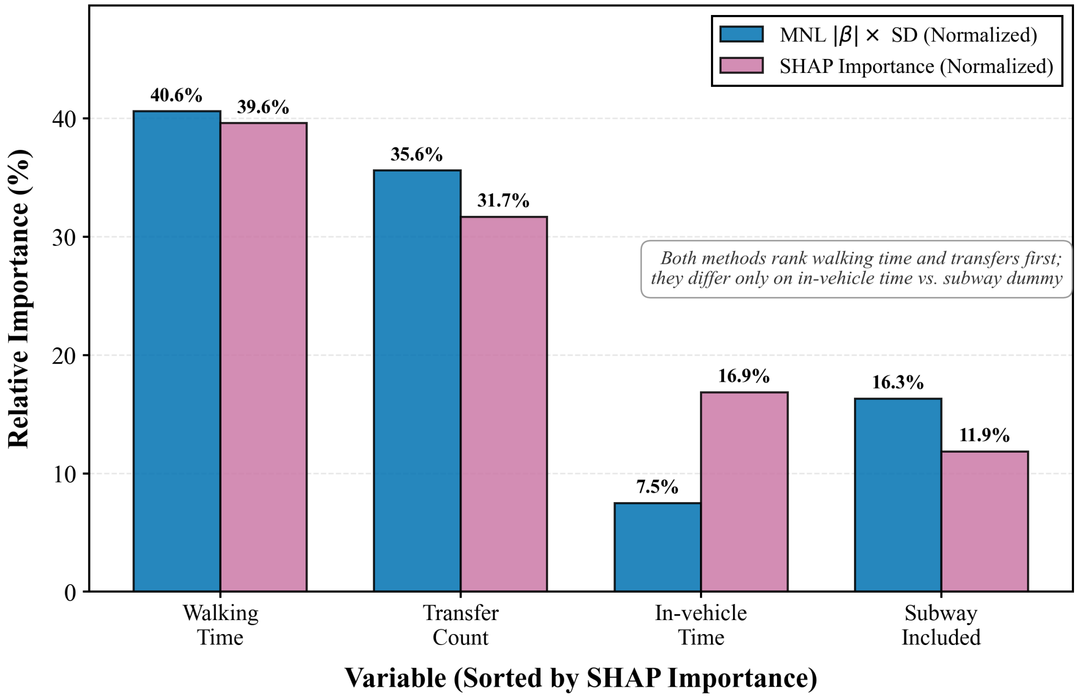

Figure 3. SHAP Variable Importance vs. Standardized MNL Coefficients ($|\beta|$·SD; Spearman ρ = 0.8)

Because the utility coefficients carry different units (per minute, per transfer, or a 0/1 dummy shift), their raw magnitudes cannot be rank-compared directly; we therefore standardize each coefficient by its attribute's test-set standard deviation ($|\beta_k| \cdot \sigma_k$), the quantity to which mean absolute SHAP contributions are proportional in a linear model. On this common scale, MNL and LightGBM agree on the two dominant factors: walking time (40.6% vs. 39.6%) and transfers (35.6% vs. 31.7%), with Spearman $\rho = 0.8$. The single rank disagreement concerns in-vehicle time and the subway dummy: T_{ride}'s per-minute coefficient is small but its variation is large (SD $\approx$ 15 min), whereas D_{subway}'s large coefficient acts on limited 0/1 variation. This agreement between the theory-driven and data-driven rankings on what dominates route choice supports our utility specification; D_{peak}'s near-zero SHAP value (0.002) further confirms that peak-hour effects do not materially shape route choice in Seoul.

ITERATIVE CALIBRATION AND MNL RE-RANKING

CALIBRATION CEILING

A natural next step is to feed estimated parameters back into OTP's cost function and iterate. Increasing walkReluctance from 1.0 to 5.0 and transferCostSeconds from 120 to 300 proved counterproductive: the Exact Match rate collapsed from 28.66% to 14.0%, because transit networks are fundamentally discrete—a modest parameter shift can cause entirely different routes to become Pareto-optimal. Holding the choice set fixed and optimizing θ solely to improve ranking yields at most +1.02%p (pooled MSA) or +0.95%p (probabilistic RSM, R^2 = 0.98). In both cases, all parameters converge to boundary values, and decomposition shows improvement is driven almost entirely by MNL re-ranking of generated alternatives, not by the router's own ranking improving. This establishes a structural ceiling of approximately +1%p for θ-based calibration.

MNL RE-RANKING: CORE CONTRIBUTION

Rather than tuning the router's cost function, we re-rank OTP's existing alternatives using estimated utility—no re-routing required:

$$\text{Rank}_{\text{MNL}} = \arg\max_j V_j = \arg\max_j \boldsymbol{\beta}'\mathbf{x}_j$$

Ranking Method	Exact Match	sim_total	N
OTP 1st Rank (gen. cost min)	20.34%	0.5372	1,320,030
MNL 1st Rank (V max)	22.23%	0.5713	1,320,030
Best Match (theoretical max)	28.66%	0.6260	1,320,030
MNL − OTP Improvement	+1.89%p	+0.034	—

Table 9. MNL Re-Ranking vs. OTP Ranking (Full Dataset, N = 1,320,030)

The gain is +1.89 percentage points on Exact Match—roughly double the calibration ceiling—at a fraction of the cost (17 minutes of scoring vs. hours of iterative OTP execution). In 75.4% of chains the two rankings agree; among the 24.6% (325,154 chains) where they diverge, MNL achieves Exact Match 10.4 times more often than OTP (27,608 vs. 2,663 chains) and outperforms on sim_total in 73.1% of disagreements (win ratio 2.7:1).

Improvement Grows with Choice Set Richness

The improvement grows monotonically with choice set size: from zero where only one alternative exists to +3.17%p with five or more alternatives, delivering its greatest value precisely where routing is most complex.

What MNL-Selected Routes Look Like

Relative to OTP's top pick, MNL favors routes with 18% less walking, 20% fewer transfers, and 2 percentage points more subway usage, at the cost of 5.9% longer ride times. Given a

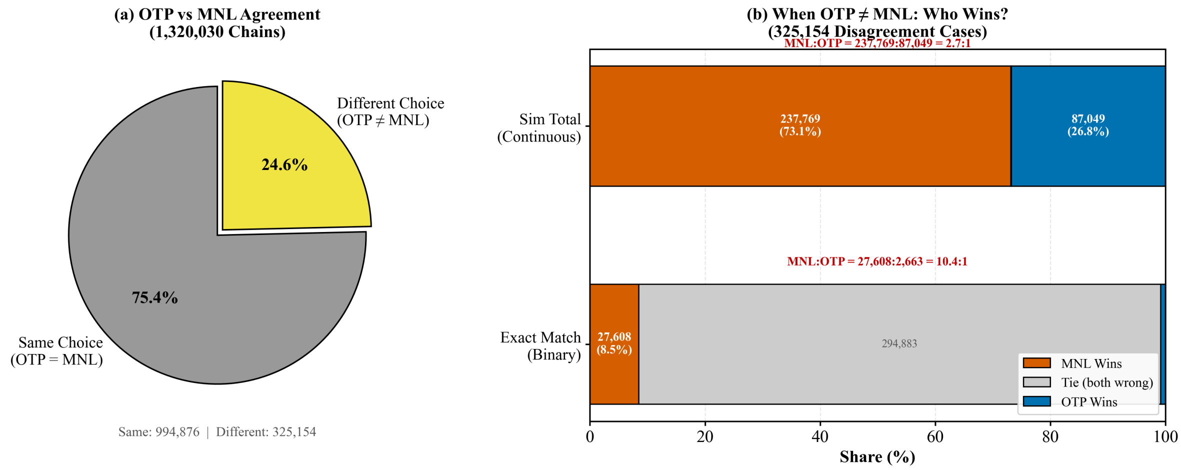

Figure 4. OTP vs. MNL Disagreement: Win Ratios by Evaluation Metric

Alternatives	Chains	OTP Exact	MNL Exact	Δ
1	345,967 (26.2%)	32.92%	32.92%	+0.00%p
2	170,755 (12.9%)	23.81%	25.33%	+1.52%p
3–4	332,660 (25.2%)	19.10%	21.33%	+2.23%p
5+	470,648 (35.7%)	10.72%	13.90%	+3.17%p

Table 10. Re-Ranking Improvement by Number of Alternatives

walking weight of 22.7×, the observed trade-off is overwhelmingly favorable from the traveler's perspective. OTP's default walkReluctance of 1.0 understates the true perceived burden by more than an order of magnitude, which is the fundamental reason its rankings diverge from observed behavior.

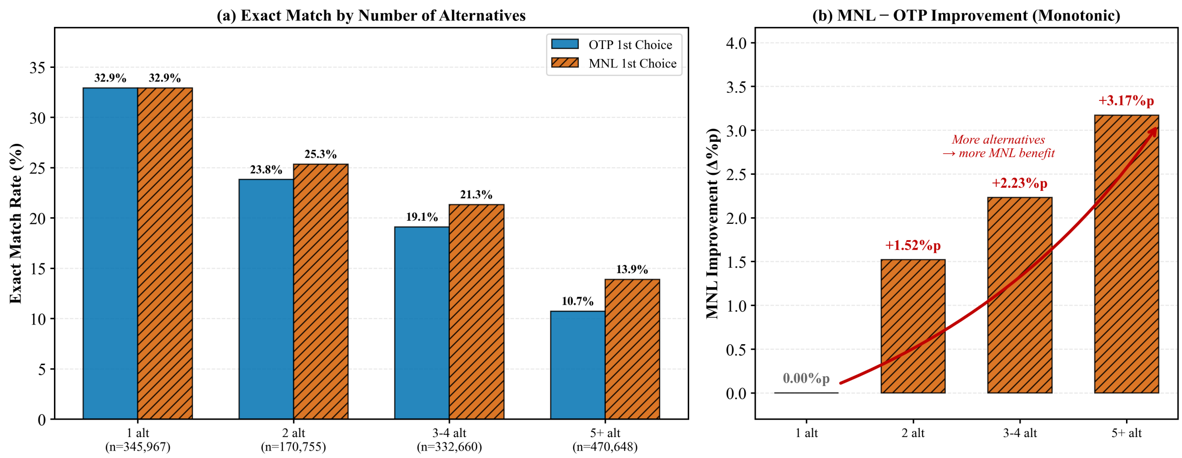

Figure 5. MNL Re-Ranking Improvement by Number of Available Alternatives

Attribute	OTP 1st Rank	MNL 1st Rank	Difference
In-vehicle time (min)	18.8	19.9	+1.1 (+5.9%)
Walking time (min)	3.2	2.6	−0.6 (−18%)
Transfers (count)	0.50	0.40	−0.10 (−20%)
Subway ratio	61.5%	63.5%	+2.0%p

Table 11. Route Characteristics: OTP vs. MNL First-Ranked Routes

CONCLUSIONS

This study pursued four objectives and offers corresponding contributions. First, the multi-level similarity framework with four hierarchical levels and ten metrics provides a systematic evaluation of route-matching quality, revealing that 28.66% of 1.32 million trip chains achieve exact match with at least one OTP alternative and that matching degrades sharply with transfer count (45.6% for direct trips vs. <0.3% for two-or-more transfers). This framework moves beyond single-metric assessments prevalent in prior work and quantifies the room for improvement that motivates the subsequent modeling. Second, the comparison of four model families on an identical hold-out set demonstrates that theory-grounded models can match or exceed machine learning: a 24-parameter Latent Class model (67.4%) outperforms LightGBM (66.2%), and a six-parameter MNL achieves 99.7% of the ML benchmark's accuracy.

Third, the LC analysis reveals that administrative card-type categories are informative but imperfect behavioral proxies (Cramér's V = 0.685): 61% of elderly users belong to the Accessibility-Priority class, but 39% exhibit Mainstream or Extreme preferences. The elderly subgroup's positive ride-time valuation and near-lexicographic transfer aversion, confirmed across all four model families, challenge the assumption that travel time minimization is universal. Fourth, MNL-based re-ranking of OTP alternatives delivers +1.89%p improvement—roughly double the structural ceiling of θ-based calibration—without re-running the router, with a 10.4:1 win ratio when the two disagree. The gain reaches +3.17%p in complex scenarios with five or more alternatives.

For transit agencies and app developers, MNL re-ranking represents a lightweight, high-impact upgrade deployable as a post-processing step with no changes to routing infrastructure. For the Accessibility-Priority segment (8.7%), systems should prioritize transfer-free subway options; for disabled users (walking weight 30.2×, transfer penalty 129.6 min), minimizing walking distance should take precedence over minimizing travel time. More broadly, card-type information should serve as a Bayesian prior for personalization, updated with revealed-preference data rather than treated as a static label.

Several limitations should be noted. The analysis draws on a single weekday, though weekday transit demand in Seoul exhibits high day-to-day regularity and 1.32 million chains far exceed typical survey-based studies. Fare differentials, real-time crowding, and weather are absent but primarily affect mode choice rather than route choice. The positive β_{ride} for elderly users invites complementary stated-preference research to disentangle seat availability from ride-time preference. Future work should extend re-ranking to full probabilistic route assignment for equity-aware service analysis, pursue real-time API deployment, multi-city replication, user-type-specific re-ranking leveraging LC class probabilities, and adaptive learning algorithms that update taste parameters as trip histories accumulate.

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